

Capital Asset Pricing Model (CAPM)

$$PV = \sum_{t=0}^{\infty} \frac{E[CF_t]}{(1+r)^t}$$

Capital Asset Pricing Model (CAPM)

- CAPM gives the expected returns of stocks trading in the market, depending on their systematic risk (beta)
- CAPM is a financial equilibrium model derived under a number of assumptions:
 - Such as: efficient markets, market in equilibrium, no-arbitrage, rational investors with diversified portfolios, linear stochastic discount factor (SDF)
- CAPM is widely used in practice for finding expected returns
 - Even if CAPM is not exactly true, it is reliable and robust

Capital Asset Pricing Model (CAPM)

- CAPM says that expected return for security i with β_i of systematic risk is:

$$r_i = r_f + \beta_i \times r_{mf}$$

- r_i is security i 's expected return
- r_f is risk-free rate, i.e., the interest rate
- r_{mf} is “market-risk-premium”
 - Typically, considered to be around 5.0% to 7.0% annually
- β_i is systematic risk

CAPM and present value (PV)

PV calculation now has two steps:

- Determine beta and use CAPM to find expected return
 - Cash flows with more systematic risk have higher expected return
- Discount future expected cash flows to calculate PV
 - Discount rate is expected return from CAPM (this is opportunity cost of capital)

$$r_i = r_f + \beta_i \times r_{mf}$$

$$PV_i = \sum_{t=1}^T \frac{E[CF_t]}{(1 + r_i)^t}$$

CAPM example

- **Example:** What is present value (PV) of the future expected cash flows ($E[CF]$): \$10, \$12, \$5, and \$10 (arriving in periods 1, 2, 3, and 4)?
- **Assumptions:**
 - Beta is 1.2
 - Risk-free rate is 4.5%
 - Market risk premium is 6.0%

CAPM and PV in Excel

	A	B	C	D	E	F	G	H
1								
2		Period	0	1	2	3	4	
3		Expected Cash flow (E[CF])	\$0.00	\$10.00	\$12.00	\$5.00	\$10.00	
4		Discounted CF	\$0.00	\$8.95	\$9.62	\$3.59	\$6.42	=G3/(1+\$C\$12)^G2
5		Present Value (PV)	\$28.58	=SUM(C4:G4)				
6								
7		Discount rate calculation						
8		Risk-free rate (r_f)	4.5%					
9		Beta	1.2					
10		Market risk premium (r_{mf})	6.0%					
11								
12		Discount rate (r)	11.7%	=C8+C9*C10				
13		($r = r_f + \text{beta} \times r_{mf}$)						

Practice Problems

Practice problems: Beta and CAPM

- **Question 1:** A project has a beta of 1.4, what is the discount rate for its cash flows?
 - **Assumption:** market risk premium is 6.0%, and risk-free rate is 4.5%
- **Question 2:** A biotech startup is working on a cure for diabetes. It has a 50-50% chance of success. If it succeeds, it will be worth billions. If it fails, it will be worthless. What is a reasonable beta and discount rate?
- **Question 3:** The US Government is issuing short-term bills to borrow money. What is a reasonable beta and discount rate for an investor in these bills?
- **Question 4:** Is it possible for a project with a low beta to have high total risk (standard deviation of returns)? Explain with an example
 - Is it possible for a project with a high beta to have a very low risk, i.e., be almost perfectly risk-free?

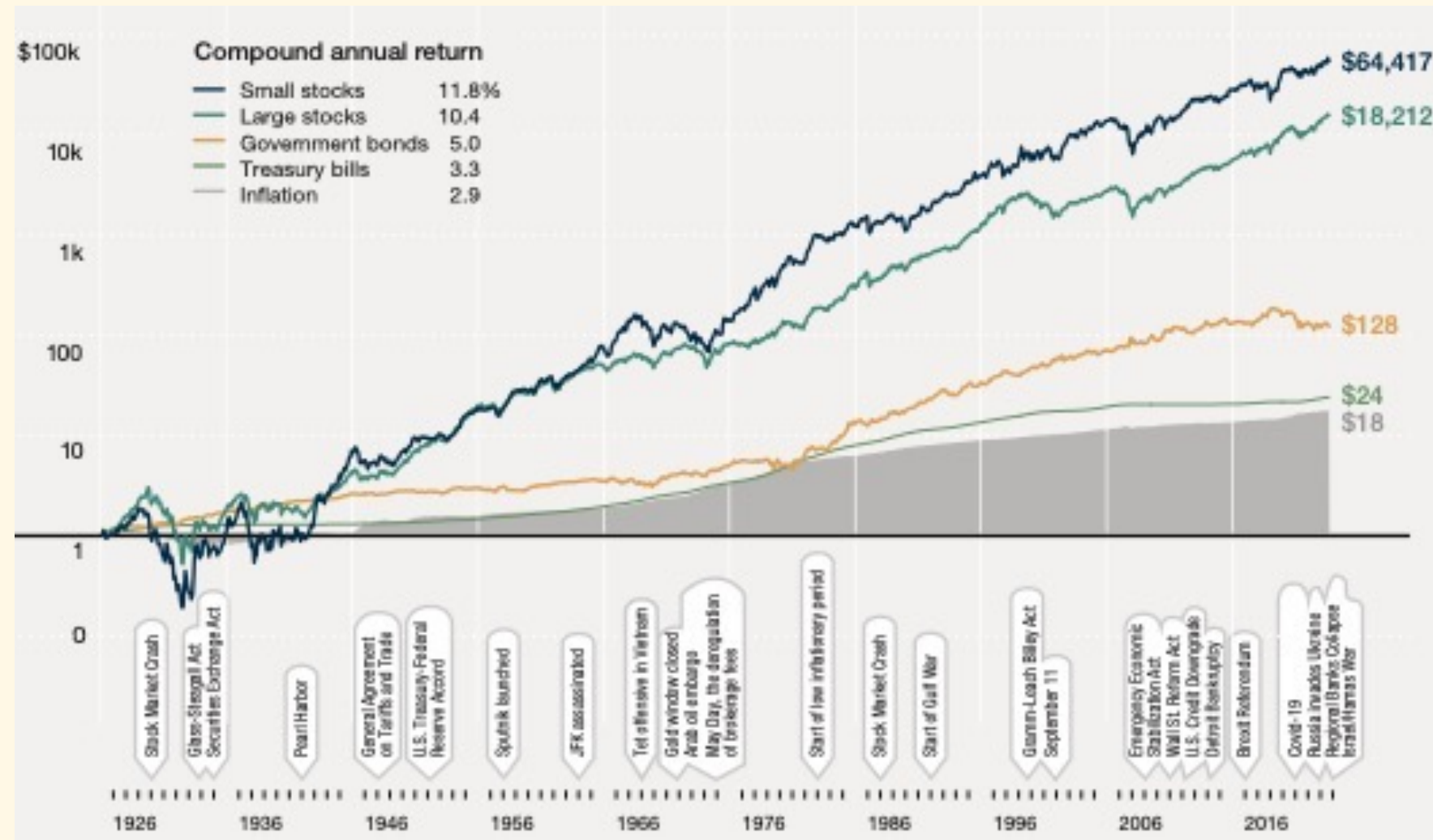
Historical Stock Returns

The Stock Market



- New York Stock Exchange (NYSE) was founded in 1792
 - World's largest stock exchange (more than USD 25T cap.)

Value of \$1 invested in 1925 through 2024



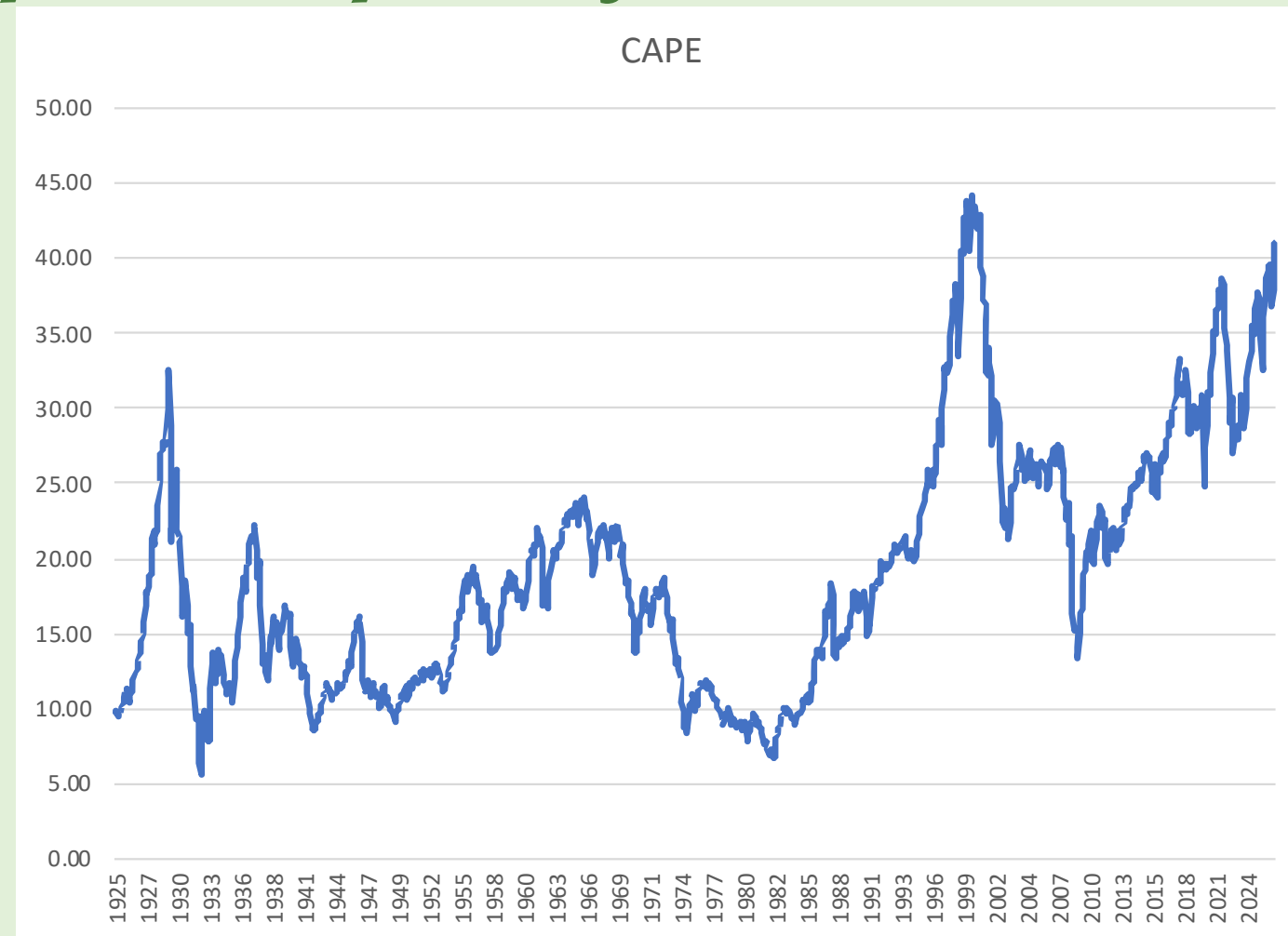
- Source: Ibbotson® SBBI® 1926-2024

Historical returns

Portfolio	Average Annual Rate of Return	Average Premium (Extra return versus Treasury bills)
Treasury bill	3.7	-
Treasury bonds	5.4	1.7
Common stocks	11.5	7.8

- Average returns on treasury bills, treasury bonds, and common stocks from 1900–2020 (in pct. per year)
 - Historically, investors have received a premium for holding risky securities (risk premium)
- Treasury bills and bonds are considered risk-free, and common stocks have earned a premium of 6.1%-7.8%
 - Exact estimate varies with the period
- This premium gives market-risk-premium in CAPM model

CAPE (Cyclically-Adjusted-Price-Earnings)



- CAPE is stock price divided by ten years of average earnings (Shiller Data, <https://shillerdata.com>)
 - Higher CAPE suggests lower future returns over 10 to 20 year horizon

Investment Decisions and Net Present Value (NPV)

Investment decisions

- A company considers investing in a project
 - Project generates a stream of cash flow
 - Goal is to maximize company's total value
 - I.e., maximize the value of the company's total cash flows
- Should we invest in project? Yes or no?
 - How to decide?

Investment decisions

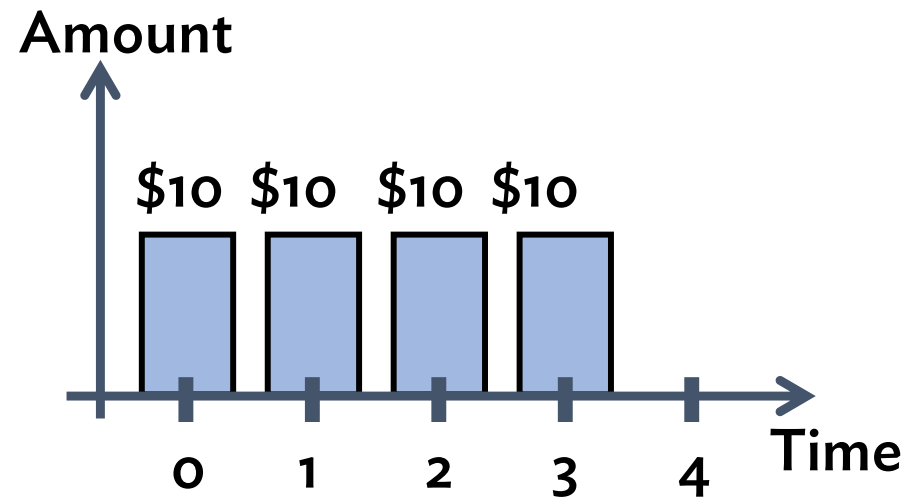
- **Conceptually:** Determine company's total cash flows (CFs) both with and without project
 - Compare the two CFs, which one is better?
 - If CF with project is better, then invest!
 - This maximizes value of company's cash flows
- **Sidenote:** In general, this works for all financial decisions

Investment decisions: Example

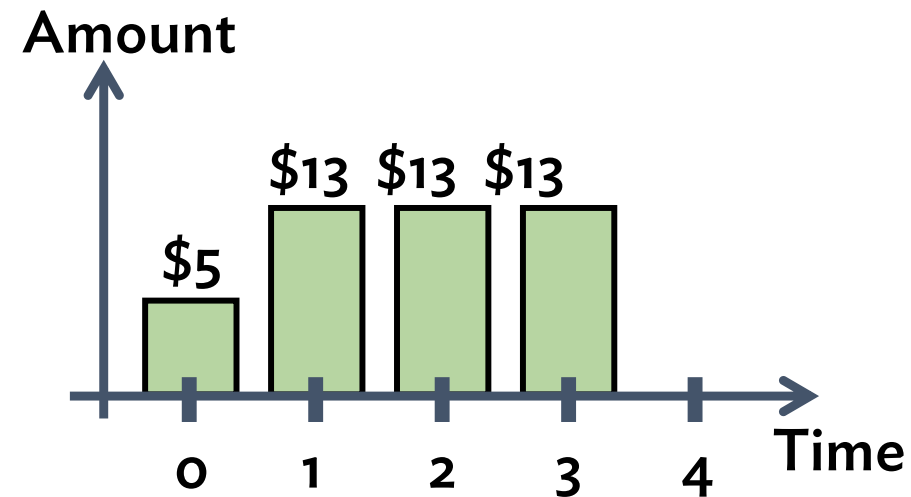
- Our company generates cash flows of \$10 per period today and for the coming three periods
- Consider investing \$5 in a new machine
 - Machine increases cash flow by \$3 per period for each of the three periods (then it is done)
- Appropriate (risk-adjusted) discount rate is 10%
- Should we invest in new machine?

Invest in new machine?

Baseline Cash flow



Cash flow with New Machine



- Without machine, PV is \$34.87
- With machine, PV is \$37.33
- Improvement from investing in machine is \$2.46 ($= \$37.33 - \34.87)
- Thus, we increase the value of our company's CFs by investing in machine!

Calculating present values (PVs)

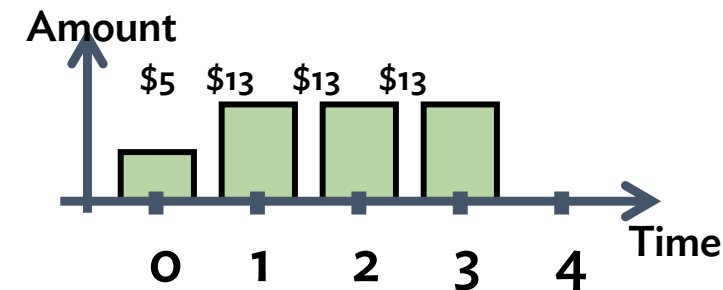
Period	0	1	2	3	4
Not Investing					
Cash flow (CF)	\$10.00	\$10.00	\$10.00	\$10.00	\$0.00
Discounted CF	\$10.00	\$9.09	\$8.26	\$7.51	\$0.00
Present Value (PV)	\$34.87				
Investing					
Cash flow (CF)	\$5.00	\$13.00	\$13.00	\$13.00	\$0.00
Discounted CF	\$5.00	\$11.82	\$10.74	\$9.77	\$0.00
Present Value (PV)	\$37.33				
Discount rate	10%				

Incremental (marginal) cash flows (CFs)

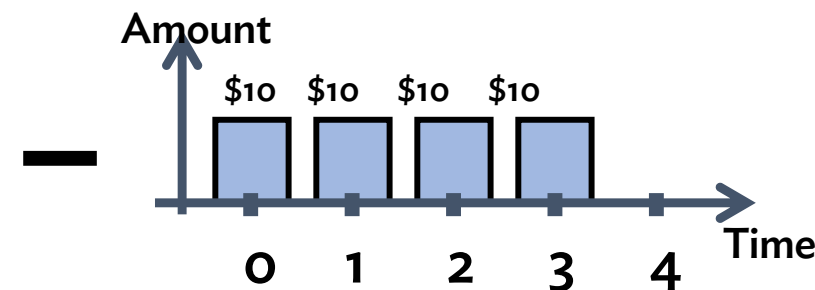
- However, figuring out a company's total cash flow is tedious
- **Trick:** Analyze investment decision from just project's incremental (marginal) cash flows (CFs)

Net Present Value (NPV)

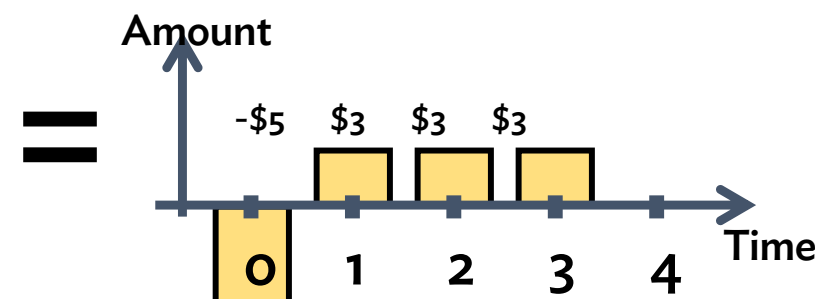
- **Net Present Value (NPV):** is the difference between PVs of company's total CFs with and without project
- Can determine NPV in two ways:
 1. Calculate PVs of each cash flow, and take difference
 2. Calculate difference in CFs and find PV of the incremental (marginal) CFs
 - Either way, NPV is \$2.46
- Investment decision can be made from incremental cash flows (CFs)



$$PV = \$37.33$$



$$PV = \$34.87$$



$$NPV = \$2.46$$

Calculating Net Present Value (NPV)

Period	0	1	2	3	4
Incremental Cash Flows (CFs)					
Cash flow (CF)	-\$5.00	\$3.00	\$3.00	\$3.00	\$0.00
Discounted CF	-\$5.00	\$2.73	\$2.48	\$2.25	\$0.00
Net Present Value (NPV)	\$2.46				
Discount rate	10%				

Net Present Value (NPV)

NPV Definition: $NPV = PV(\text{CFs with project}) - PV(\text{CFs without project})$

- **$NPV = PV(\text{incremental CFs})$**

NPV Rule: Invest when $NPV > 0$

NPV alternative definition

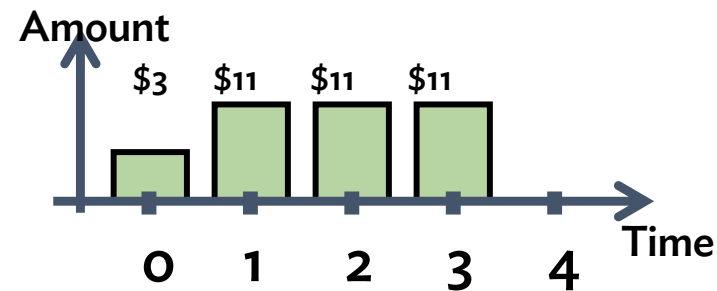
- Net present value (NPV) is sometimes stated as: **$NPV = PV - I$**
 - Where **I** is initial investment in project, and **PV** is PV of project's future cash flow
- Our definition is: **$NPV = PV(\text{project's incremental CF})$**
 - The two definitions are equivalent for projects with just a single initial investment (outflow)
 - Our definition is better, and more general. It is a reminder that objective is to increase value of firm's total CFs
 - Helps avoid two pitfalls: **(1)** Opportunity costs, **(2)** sunk costs

Pitfall 1: Opportunity cost

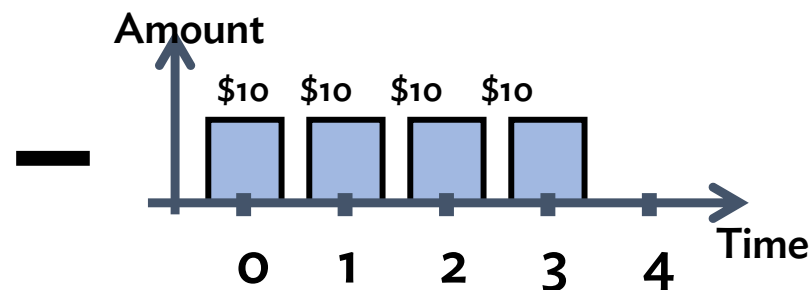
- Back to example, new machine still costs \$5 and still generates \$3 worth of extra sales each period
- However, it will be put in our empty production hall
 - Instead, we would rent out this hall for \$2 per period
 - This rental income is included in the \$10 per period
- How does that affect NPV calculation?

Opportunity cost and NPV

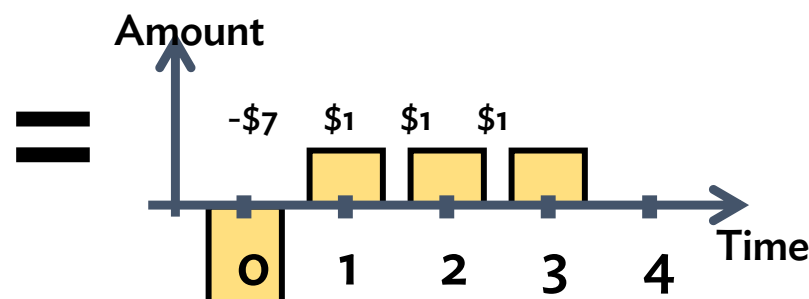
- Investing in new machine reduces rental income
 - Top CF adjusted accordingly
 - Lost rental income is not a direct cost of new machine
- Now NPV is negative
 - Shouldn't invest
- Thinking in terms of total CFs gets this right



$$PV = \$30.36$$



$$PV = \$34.87$$



$$NPV = -\$4.51$$

Pitfall 2: Sunk cost fallacy

- Sunk-cost fallacy is the error of continuing a project or investment based on the amount already spent, rather than based on future costs and benefits
 - Rational decision-making ignores past, unrecoverable expenditures (sunk costs)
- **Example:** A company has invested \$5 million in developing a new product, but market research now shows weak demand and high competition.
 - Sunk-cost fallacy would be to complete the project to avoid “wasting” product development investment
 - Rational decisions should only consider whether continuing is likely to generate a positive return
- Focusing on the difference in total future cash flows (incremental CF) avoids this mistake
- Sunk costs are not marginal